

# CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

2<sup>nd</sup> Semester of B. Pharm Examination

University Theory Examination October/November-2016

MA 141 Advanced Mathematics

Date: 09.11.2016, Wednesday

Time: 10:00a.m to 01:00p.m

Maximum Marks: 80

## Instruction:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

## SECTION-I

### Q-1. Attempt any TWO of the following :

- A
- (i) Find 5<sup>th</sup> derivative of  $e^{2x}$ . [02]
- (ii) Find inverse Laplace transform of the following function [04]
- $$(a)f(s) = \frac{3}{s^2 + 9} \quad (b)f(s) = \frac{s}{s^2 + 1}$$
- (iii) Using Leibnitz's theorem find  $(x \cos 2x)_n$ . [04]
- B
- (i) Find Laplace transform of the function  $e^{2t} + 3 \sin 4t$  [02]
- (ii) Verify Lagrange's mean value theorem for  $f(x) = x^2$ ,  $x \in [0, 5]$  [04]
- (iii) Write Maclaurin's series expansion of the function  $f(x) = e^x$ . [04]
- C
- (i) State Cauchy's mean value theorem. [02]
- (ii) Find Taylor's series expansion for the function  $x^2 + x + 1$  around  $a=5$ . [04]
- (iii) Using Laplace transformations solve  $y'' + y = 0$ ,  $y(0) = 0$ ,  $y'(0) = 1$  [04]

### Q-2. Attempt any TWO of the following :

- A
- (i) State Leibnitz's theorem. [02]
- (ii) Verify Cauchy's mean value theorem for the functions  $f(x) = x^2$  and  $g(x) = x^3$  in  $[1, 2]$ . [04]
- (iii) Find the 3<sup>rd</sup> derivative of the function  $\cos(2x + 7)$  [04]
- B
- (i) If  $y(x) = e^{3x} \sin(7x + 4)$  then  $y_n =$  \_\_\_\_\_. [02]
- (ii) Verify Lagrange's mean value theorem for,  $x^2 + 5x + 1$ ,  $x \in [-2, 2]$  [04]
- (iii) Find  $L\left(\sin 3t - 7t^3 + \frac{1}{5} \cosh 3t + 2e^{-t}\right)$  [04]
- C
- (i) Find (a)  $L(\sin 6t)$  (b)  $L(t^2)$  [02]
- (ii) Using Leibnitz's theorem, find the  $n^{\text{th}}$  derivative of the function  $x^3 \sin 2x$ . [04]
- (iii) Write Maclaurin's series expansion of  $f(x) = \cosh x$ . [04]

## SECTION-II

**Q-3. Attempt any TWO of the following :**

- (i) Find order and degree of  $\left(\frac{d^3y}{dx^3}\right)^5 + 4\frac{dy}{dx} = 2x$

- (ii) Find general solution of  $y'' + 4y' + 4y = 0$

- (iii) Suppose that in the course of any given year the number of cases of a disease is reduced by 20 percentage. If there are 10,000 cases today, how many years it will take to reduce the number to 1000?

**B** (i) Write the general form of Linear equations of first order and first degree.

- (ii) Solve the Linear differential equation  $\frac{dy}{dx} + 2y = e^{-2x}$ .

- (iii) Show that  $y = ae^{-x} + be^{6x}$  is a solution of  $y'' - 5y' - 6y = 0$ .

**C** (i) Write the characteristic equation for  $ay'' + by' + cy = 0$

- (ii) Show that  $y = ax^3 + bx^2$  is a solution of  $x^2y'' - 4xy' + 6y = 0$ .

- (iii) Solve by the method of variation of parameters  $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = e^{3x}$

**Q-4. Attempt any TWO of the following :**

- (i) Find general solution of  $y'' + y = 0$

- (ii) Solve the Linear differential equation  $\frac{dy}{dx} - 2\frac{x}{y} = x^3$ .

- (iii) Find general solution of (a)  $y'' + y' + 2y = 0$  (b)  $y'' + 2y' - 3y = 0$

- (i) Find order and degree of  $\left(\frac{d^7y}{dx^7}\right)^3 = \sin 3x + \left(\frac{d^2y}{dx^2}\right)^3$

- (ii) Solve Exact differential equation  $(y + e^{-x})dx + (x + 5)dy = 0$

- (iii) In a city, rate of increase in population is proportional to the size of the population, if the population of the city in 1960 is 1,25,000 and 1,40,000 in 1975, what will be its population in 2000?

**C** (i) Solve Exact differential equation  $(x^2 - y^2)dx = 2xydy$

- (ii) Find general solution of (a)  $y'' + 22y' + 121y = 0$  (b)  $y'' + 12y' - 45y = 0$

- (iii) Solve the Linear differential equation  $x\frac{dy}{dx} + y = e^{-x}$ .

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